

I: Omens

Source file name: `omens.c`, `omens.cpp`, `omens.java`, or `omens.py`

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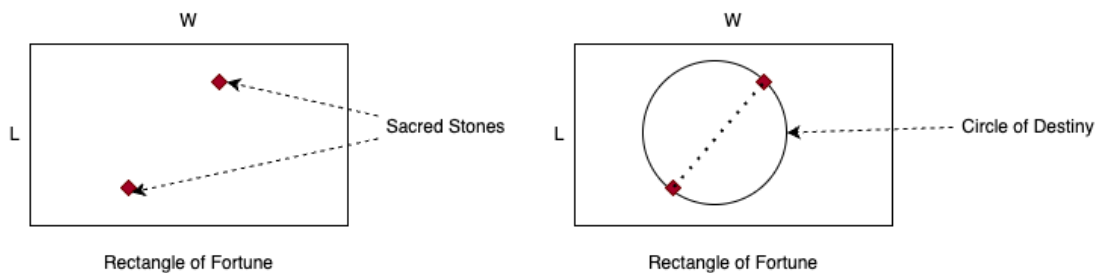
Hanna Lonely, captain of the starship Century Hawk, and her crew, have arrived to the planet C0-3, carrying precious (and dubiously legal) cargo for the Kingdom of Fomana.

The Fomanians are a very superstitious people. Once a year (that's what they call their cycle around their star, CO), they celebrate the *Omens Day*, which happens to be the next day Captain Lonely and her crew arrived to C0-3. On this day, the Royal Oracle performs the divination ritual, by throwing a pair of *Sacred Stones* on the *Rectangle of Fortune*.

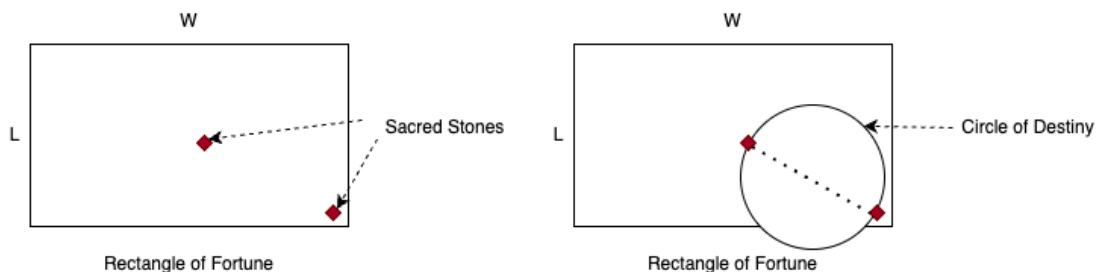
The *Rectangle of Fortune* is an $L \times W$ rectangle, precisely drawn on the sands of the *Sacred Beach*. Once the Royal Oracle has thrown the *Sacred Stones*, she proceeds to draw the *Circle of Destiny*, in which the stones are in exactly opposite sides; the diameter of the circle is the distance between the two stones.

Fomanians believe it's a *good omen*, when the resulting *Circle of Destiny* is completely contained within the *Rectangle of Fortune*, clearly meaning their destiny in good fortune. On the other hand, a *bad omen* is when the *Circle of Destiny* is partially outside the *Rectangle of Fortune*.

This is an example of *good omen*:



This is an example of *bad omen*:



The nature of the *Sacred Stones* is to always fall within the *Rectangle of Fortune*, and you can assume they are equally likely to fall anywhere within the rectangle.

The recently appointed *Royal Oracle* is quite progressive. She went to the Galactic University, and wants Captain Lonely to help her calculate the probability a thrown resulting in a good omen, given the dimensions of the *Rectangle of Fortune*. As the most junior member of the crew, you are tasked with this calculation.

Input

The input consists of several test cases. A case is defined with a line with two positive integer values L and W , $1 \leq L, W \leq 1\,000$, representing the dimensions of the *Rectangle of Fortune*.

L W

The end of the input is signaled with a line

0 0

that should be not processed.

The input must be read from standard input.

Output

For each test case output a line with the probability of the *Circle of Destiny* being completely contained within the *Rectangle of Fortune*, rounded to 4 decimal places.

The output must be written to standard output.

Sample Input	Sample Output
1 1	0.5236
20 40	0.3927
40 30	0.4909
0 0	